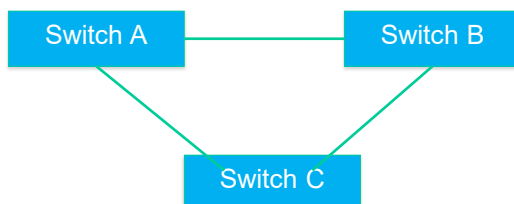


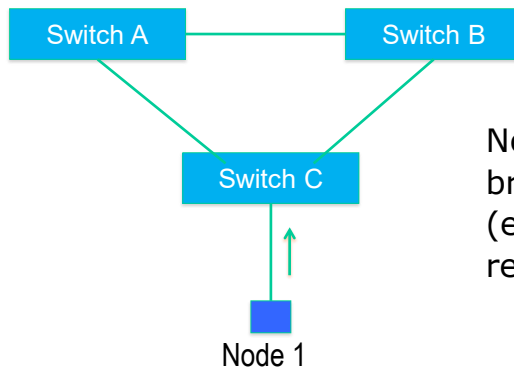
Switching Loops

Switching Loop

- When there is more than one path between two switches
- What are the potential problems?

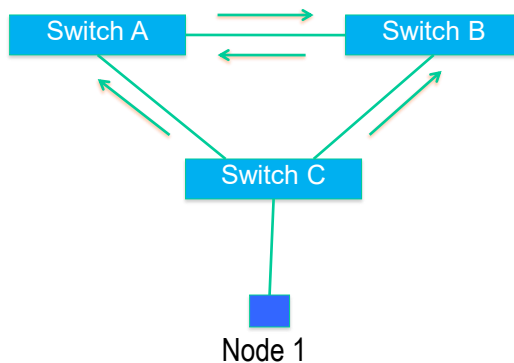


Switching Loop



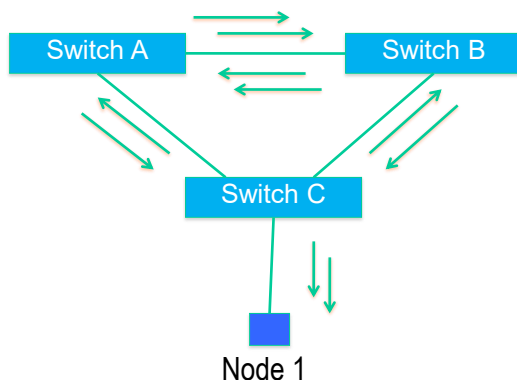
Node1 sends a broadcast frame (e.g. an ARP request)

Switching Loop



Switches A, B and C broadcast node 1's frame out every port

Broadcast Storm



But they receive each other's broadcasts, which they need to forward again out every port!

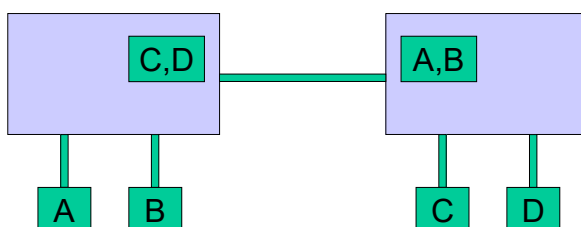
The broadcasts are amplified, creating a ***broadcast storm!***

Broadcast storm

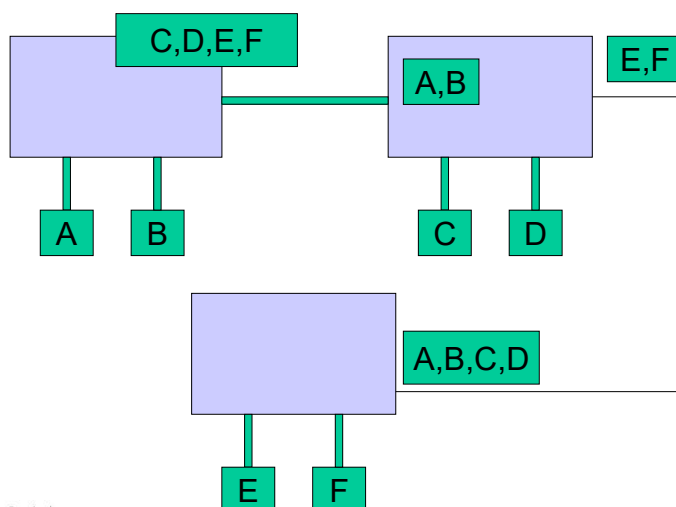
- Broadcast traffic is carried across all trunks in a VLAN.
- A truck link carries broadcast traffic from several VLANs.
- Broadcast storm can spread across the extent of the VLAN.
- Maintenance and troubleshooting nightmare.

Switching Loop

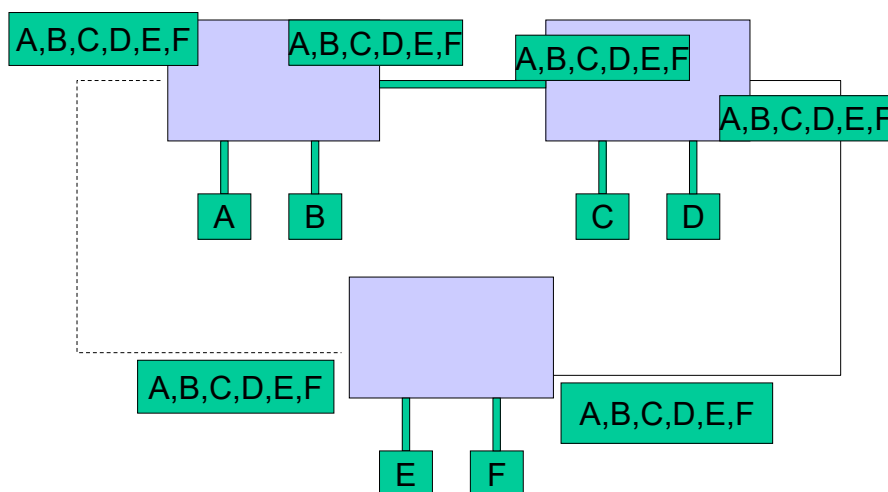
- Os switches criam tabelas de encaminhamento escutando os endereços MAC de origem enviados para as suas portas.



Cascadeamento de Switches



Cascadeamento de Switches



Switching Loops

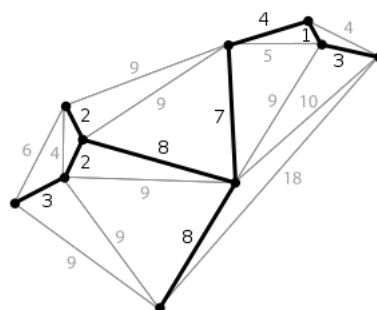
- If there is more than one path between two switches:
 - **Forwarding tables become unstable**
 - Source MAC addresses are repeatedly seen coming from different ports
 - **Switches will broadcast each other's broadcasts**
 - All available bandwidth is utilized
 - Switch processors cannot handle the load

Good Switching Loops???

- But you can take advantage of loops!
 - **Redundant paths improve resilience when:**
 - A switch fails
 - Wiring breaks
- How to achieve redundancy without creating dangerous traffic loops?
- Solution: **Spanning Tree Protocol**

What is a Spanning Tree

- “Given a connected, undirected graph, a spanning tree of that graph is a subgraph which is a tree and connects all the vertices together”.
- A single graph can have many different spanning trees.



STP: Spanning-Tree Protocol

- O STP é executado em cada switch da rede.
- Princípio:
 - Entre 2 estações na rede só pode existir um caminho ativo.
 - As portas que impliquem loops fechados devem ser bloqueadas.
- A estratégia consiste em escolher um Switch como Root, e construir uma árvore com o menor caminho até ao Root.

STP

- O STP utiliza mensagens BPDU:
 - Bridge Protocol Data Unit
 - Mensagens em Multicast (MAC)
 - DE: 01:80:C2:00:00:00
 - ATÉ: 01:80:C2:00:00:01
- STP funciona continuamente, de maneira a refletir mudanças de topologia na rede.
 - Se SPT está ativo, os pacotes BPDU são recebidos, mas não encaminhados.
 - Se SPT está desativado, os pacotes BPDU são encaminhados como pacotes multicast genéricos.

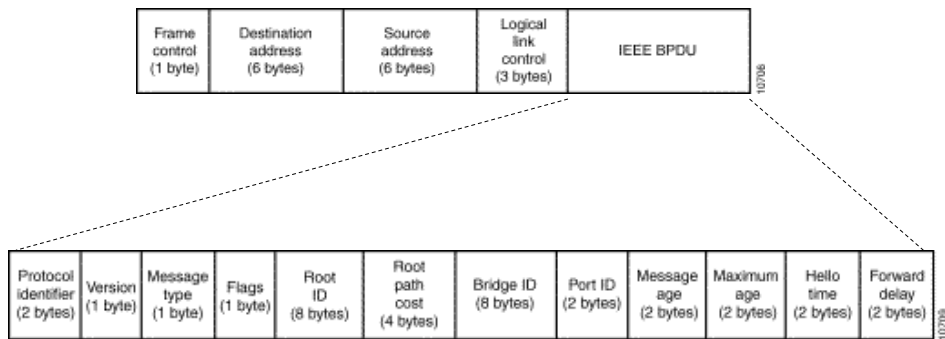
Spanning Tree Protocol

- Several flavors:
 - Traditional Spanning Tree (802.1d)
 - Rapid Spanning Tree or RSTP (802.1w)
 - Multiple Spanning Tree or MSTP (802.1s)

Traditional Spanning Tree (802.1d)

- Switches exchange messages that allow them to compute the Spanning Tree
 - These messages are called BPDUs (Bridge Protocol Data Units)
 - Two types of BPDUs:
 - Configuration
 - Topology Change Notification (TCN)

BPDUs: Padrão IEEE 802.1D



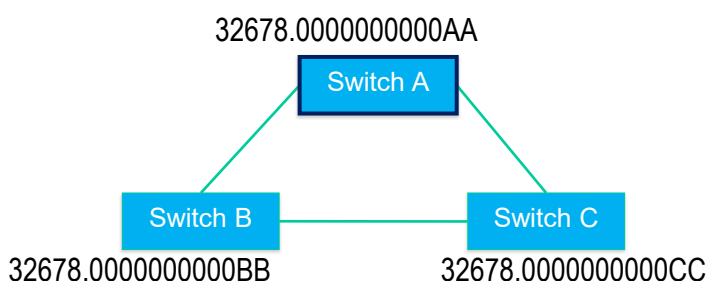
Traditional STP (802.1d)

- First Step:
 - Decide on a point of reference: the Root Bridge
 - The election process is based on the Bridge ID, which is composed of:
 - The Bridge Priority: A two-byte value that is configurable
 - The MAC address: A unique, hardcoded address that cannot be changed.

Root Bridge Selection (802.1d)

- Each switch starts by sending out BPDUs with a Root Bridge ID equal to its own Bridge ID.
 - **I am the root!**
- Received BPDUs are analyzed to see if a lower Root Bridge ID is being announced.
 - **If so, each switch replaces the value of the advertised Root Bridge ID with this new lower ID.**
- Eventually, they all agree on who the Root Bridge is.

Root Bridge Selection (802.1d)



- All switches have the same priority.
- Switch A has the lowest Id.
- Switch A becomes the Root.

Root Port Selection (802.1d)

- Now each switch needs to figure out where it is in relation to the Root Bridge
 - Each switch needs to determine its Root Port
 - The key is to find the port with the lowest Root Path Cost
 - The cumulative cost of all the links leading to the Root Bridge

Root Port Selection (802.1d)

- Each link on a switch has a Path Cost
 - Inversely proportional to the link speed
e.g. the faster the link, the lower the cost

Link Speed	STP Cost
10 Mbps	100
100 Mbps	19
1 Gbps	4
10 Gbps	2

Root Port Selection (802.1d)

- Root Path Cost is the accumulation of a link's Path Cost and the Path Costs learned from neighboring Switches.
 - **It answers the question: How much does it cost to reach the Root Bridge through this port?**

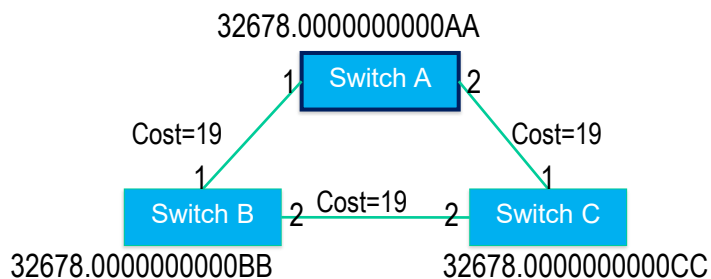
Root Port Selection (802.1d)

- Root Bridge sends out BPDUs with a Root Path Cost value of 0
- Neighbor receives BPDU and adds port's Path Cost to Root Path Cost received
- Neighbor sends out BPDUs with new cumulative value as Root Path Cost
- Other neighbor's down the line keep adding in the same fashion

Root Port Selection (802.1d)

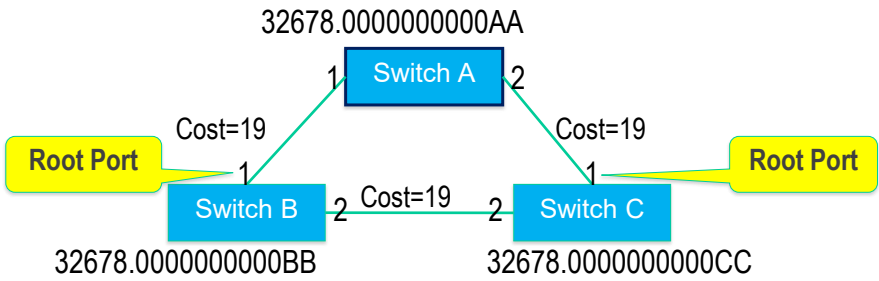
- On each switch, the port where the lowest Root Path Cost was received becomes the **Root Port**
 - This is the port with the best path to the Root Bridge

Root Port Selection (802.1d)

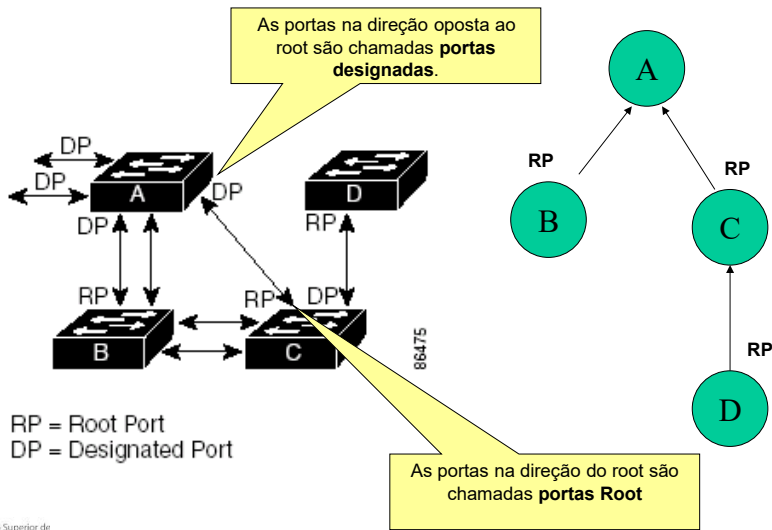


- What is the Path Cost on each Port?
- What is the Root Port on each switch?

Root Port Selection (802.1d)



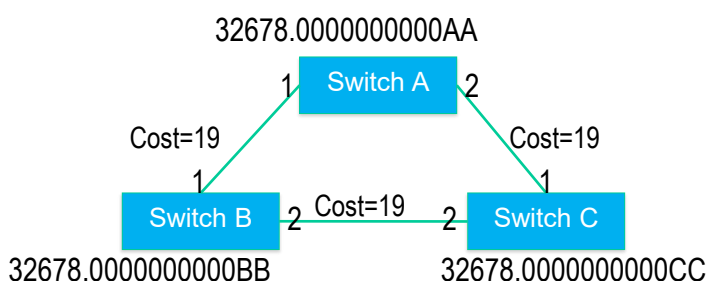
Topologia STP



Electing Designated Ports (802.1d)

- OK, we now have selected root ports but we haven't solved the loop problem yet, have we? **The links are still active!**
- Each network segment needs to have only one switch forwarding traffic to and from that segment
- Switches then need to identify one **Designated Port** per link
 - **The one with the lowest cumulative Root Path Cost to the Root Bridge**

Designated Port Selection (802.1d)

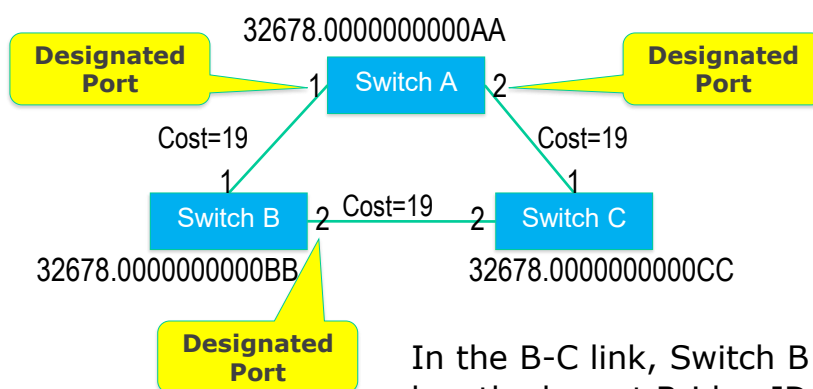


Which port should be the **Designated Port** on each **segment**?

Electing Designated Ports (802.1d)

- Two or more ports in a segment having identical Root Path Costs is possible, which results in a tie condition
- All STP decisions are based on the following sequence of conditions:
 - **Lowest Root Bridge ID**
 - **Lowest Root Path Cost to Root Bridge**
 - **Lowest Sender Bridge ID**
 - **Lowest Sender Port ID**

Root Port Selection (802.1d)

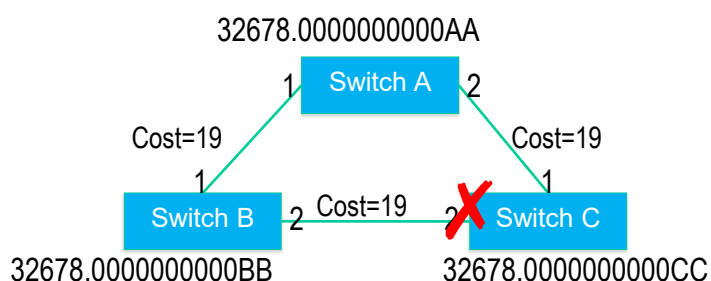


In the B-C link, Switch B has the lowest Bridge ID, so port 2 in Switch B is the Designated Port

Blocking a port

- Any port that is not elected as either a Root Port, nor a Designated Port is put into the Blocking State.
- This step effectively breaks the loop and completes the Spanning Tree.

Blocking a port



Port 2 in Switch C is put into the **Blocking State**, because it is **neither a Root Port nor a Designated Port**

Spanning Tree Protocol States

- Disabled
 - Port is shut down
- Blocking
 - Not forwarding frames
 - Receiving BPDUs
- Listening
 - Not forwarding frames
 - Sending and receiving BPDUs

Spanning Tree Protocol States

- Learning
 - Not forwarding frames
 - Sending and receiving BPDUs
 - Learning new MAC addresses
- Forwarding
 - Forwarding frames
 - Sending and receiving BPDUs
 - Learning new MAC addresses

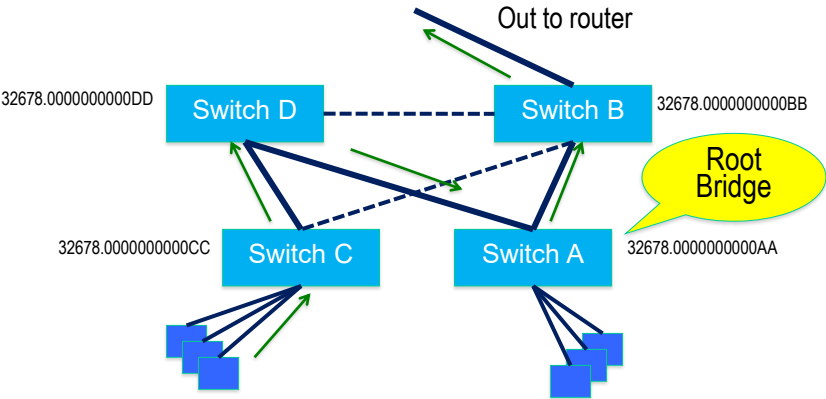
STP Topology Changes

- Switches will recalculate if:
 - **A new switch is introduced**
 - It could be the new Root Bridge!
 - **A switch fails**
 - **A link fails**

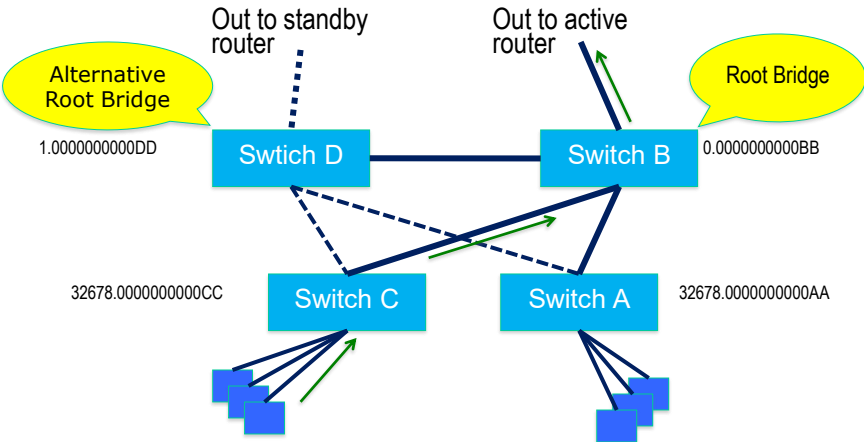
Root Bridge Placement

- Using default STP parameters might result in an undesired situation
 - **Traffic will flow in non-optimal ways**
 - **An unstable or slow switch might become the root**
- You need to plan your assignment of bridge priorities carefully

Bad Root Bridge Placement



Good Root Bridge Placement



Protecting the STP Topology

- Some vendors have included features that protect the STP topology:
 - **Root Guard**
 - **BPDU Guard**
 - **Loop Guard**
 - **UDLD**
 - **Etc.**

STP Design Guidelines

- Enable spanning tree even if you don't have redundant paths
- Always plan and set bridge priorities
 - **Make the root choice deterministic**
 - **Include an alternative root bridge**
- If possible, do not accept BPDUs on end user ports

802.1d Convergence Speeds

- Moving from the Blocking state to the Forwarding State takes at least 2 x Forward Delay time units (~ 30 secs.)
 - **This can be annoying when connecting end user stations**
- Some vendors have added enhancements such as PortFast, which will reduce this time to a minimum for edge ports
 - **Never use PortFast or similar in switch-to-switch links**
- Topology changes typically take 30 seconds too
 - **This can be unacceptable in a production network**

Rapid Spanning Tree (802.1w)

- Convergence is much faster
 - **Communication between switches is more interactive**
- Edge ports don't participate
 - **Edge ports transition to forwarding state immediately**
 - **If BPDUs are received on an edge port, it becomes a non-edge port to prevent loops**

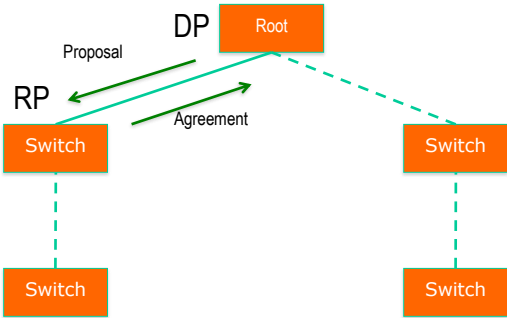
Rapid Spanning Tree (802.1w)

- Defines these port roles:
 - **Root Port** (same as with 802.1d)
 - **Alternate Port**
 - A port with an alternate path to the root
 - **Designated Port** (same as with 802.1d)
 - **Backup Port**
 - A backup/redundant path to a segment where another bridge port already connects.

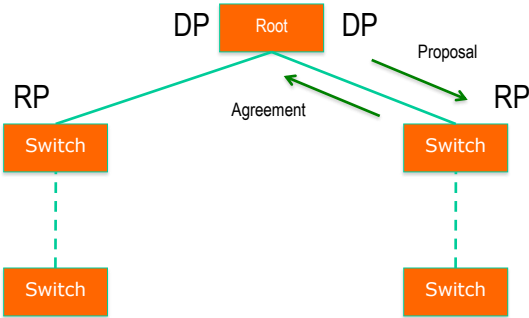
Rapid Spanning Tree (802.1w)

- Synchronization process uses a handshake method
 - **After a root is elected, the topology is built in cascade, where each switch proposes to be the designated bridge for each point-to-point link**
 - **While this happens, all the downstream switch links are blocking**

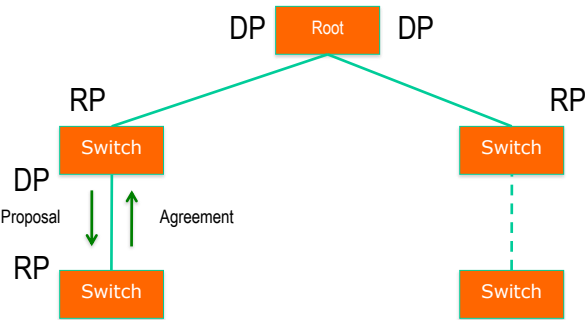
Rapid Spanning Tree (802.1w)



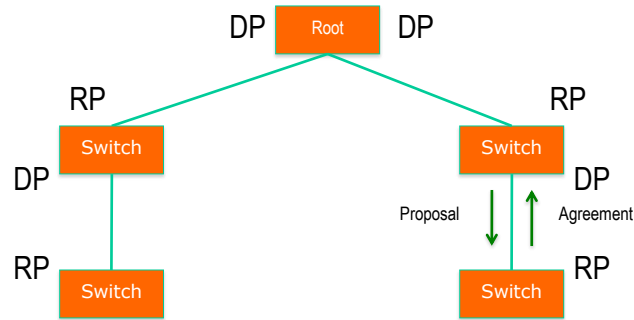
Rapid Spanning Tree (802.1w)



Rapid Spanning Tree (802.1w)



Rapid Spanning Tree (802.1w)



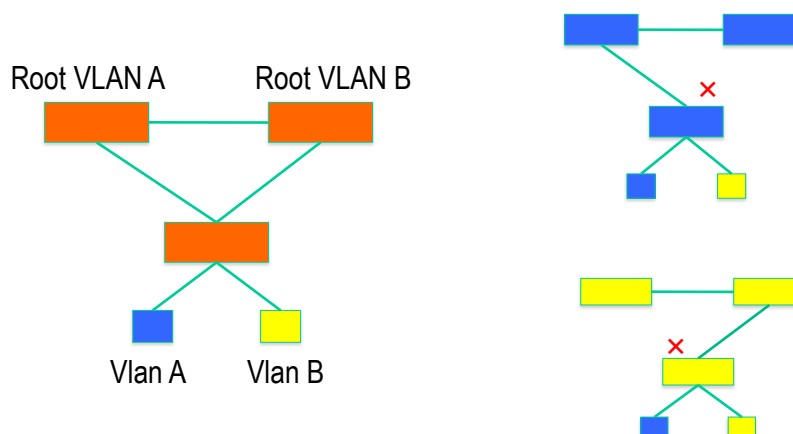
Rapid Spanning Tree (802.1w)

- Prefer RSTP over STP if you want faster convergence.
- Always define which ports are edge ports.

Multiple Spanning Tree (802.1s)

- Allows separate spanning trees per VLAN group
 - **Different topologies allow for load balancing between links**
 - **Each group of VLANs are assigned to an “instance” of MST**
- Compatible with STP and RSTP

Multiple Spanning Tree (802.1s)



VLANS increase complexity

- You can no longer “just replace” a switch
 - **Now you have VLAN configuration to maintain**
 - **Field technicians need more skills**
- You have to make sure that all the switch-to-switch trunks are carrying all the necessary VLANs
 - **Need to keep in mind when adding/removing VLANs**